

FACTSHEET FS10 · Betting against beta

What changed on 2 October 2026. Two corrections, both to the universe of stocks, and every number below is re-run on them. (1) **US:** the universe is now the actual S&P 500 members at each month-end. Until now it was the 500 largest of every company that was *ever* in the index 2012–2026, which includes later winners before they joined. (2) **UK and EU:** the price panels now include the index members that stopped trading (UK coverage of members 71% → 92%, EU 81% → 89%); before, most of them had no prices. SCANDI stays as registered (OMXC25 + OMXS30 + OMXH25). Registered verdicts are unchanged; where a corrected number crosses a gate, the text says so. Comparison and method: FACTSHEETS_RERUN_2026-10-02.md.

US, 1999–2012 (pre-registered second period, run 2 October 2026). The same rules on Sharadar's point-in-time S&P 500 with delisted stocks, 1999–2012, a period this factsheet was not built on. L/S net: t +0.39 (2013–2026: +1.17), same sign, weaker. Long-only alpha vs the equal-weight universe: +2.4% a year, t +1.41 (2013–2026: +1.93), same sign, weaker. Registered verdicts are unchanged. Pre-registration, method and every result: PREREG_US_1999_2012.md.

Lever up the low-beta stocks, short the high-beta ones

Henrik Gjerming · Rude Investment Consulting · Project 10 sidebar: strategy factsheets · data to 31 August 2026

STRATEGY	ORIGIN	FAMILY
Defensive / betting against beta	Frazzini & Pedersen (2014)	Low risk
UNIVERSES	BACKTEST	BOOKS
US · EU · UK · DK · SCANDI · World (point-in-time)	Feb 2013 – Aug 2026, monthly, net of costs	Long/short and long-only (low-beta stocks)
REBALANCE	STATUS	
Monthly	Descriptive factsheet, not a registered trial	

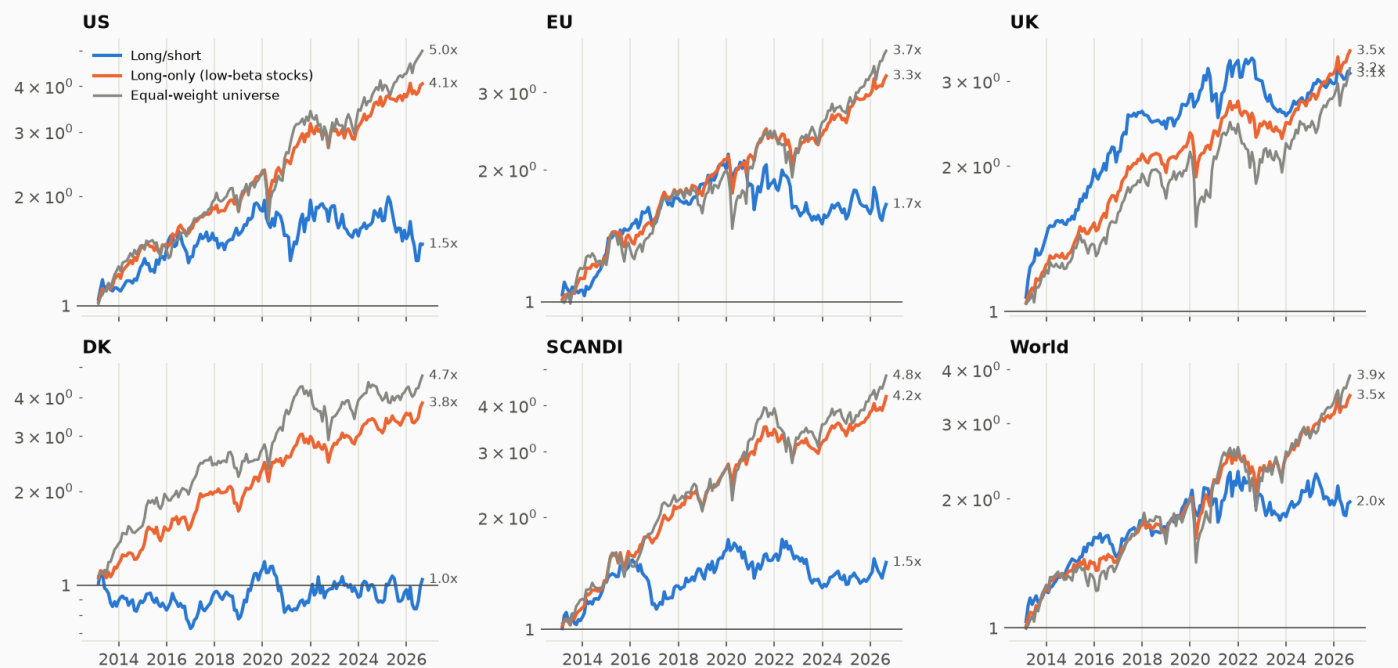
The claim. Low-beta stocks have higher risk-adjusted returns than high-beta stocks. A portfolio that is long leveraged low-beta stocks and short de-leveraged high-beta stocks, beta-neutral, earned a large positive return in the US 1926–2012 and in 19 other markets (Frazzini & Pedersen 2014). **What we find, 2013–2026, point-in-time universes, after costs:** the long/short earned +1.7% to +9.1% a year and was positive in all six; no universe passes the 2.87 gate. The long-only book (low-beta stocks) had a higher Sharpe than the equal-weight universe in all six, with alphas of +2.4% to +3.7%, passing the gate in UK. **What goes wrong (§10):** the main drags are a weaker second half (2020–26). None of the pre-declared fixes passes the gate in both windows. **Classification (§11):** Low risk; its returns sit closest to the low risk cluster of the signal library. **360° view (§12):** nearest neighbours Volatility (252 days), Volatility; after its closest neighbours and the market it keeps an alpha of -5.9% to +4.3% (largest |t| 2.2), so it carries some information of its own.

1. Headline performance

Universe	Names / groups	L/S return / yr	L/S t	L/S CAPM alpha (t)	L/S beta	Long-only CAGR	Long-only Sharpe	Long-only max DD	EW CAGR	EW Sharpe	Size-weighted CAGR
US	502 / 10	+4.3%	1.17	+2.4% (0.6)	0.14	+10.9%	0.94	-20%	+12.6%	0.85	+14.9%
EU	264 / 10	+4.6%	1.59	+3.6% (1.2)	0.09	+9.1%	0.91	-18%	+10.2%	0.75	+10.2%
UK	326 / 10	+9.1%	2.80	+7.7% (2.5)	0.14	+9.6%	1.01	-19%	+8.9%	0.68	+7.8%
DK	20 / 3	+1.7%	0.39	+1.1% (0.3)	0.04	+10.4%	0.87	-19%	+12.0%	0.83	+10.2%
SCANDI	70 / 5	+3.8%	1.23	+3.3% (1.1)	0.04	+11.2%	1.05	-18%	+12.3%	0.89	+11.0%
World	1088 / 10	+5.9%	2.12	+3.2% (1.1)	0.24	+9.6%	0.85	-21%	+10.5%	0.71	+10.3%

Monthly, local currency (World in USD). L/S = equal-weighted low-beta stocks minus high-beta stocks (rank-weighted, each leg scaled to beta 1), net of 10 bp per side on turnover, a size-tiered borrow fee on the short leg and, where the book is net long cash, financing at the USD risk-free rate + 0.5%. Newey-West t (6 lags). CAPM vs the equal-weight universe. Size-weighted universe = market-cap weights in the US, traded-value weights elsewhere (a proxy for the index).

Betting against beta on six stock universes, 2013-2026



Growth of 1, log scale, local currency, net of costs. Henrik Gjering · Project 10 factsheets

2. Strategy description

Low-beta stocks have higher risk-adjusted returns than high-beta stocks. A portfolio that is long leveraged low-beta stocks and short de-leveraged high-beta stocks, beta-neutral, earned a large positive return in the US 1926–2012 and in 19 other markets (Frazzini & Pedersen 2014).

Why it might work. Leverage-constrained investors buy high-beta stocks instead of levering the market, bidding them up (Black 1972; Frazzini & Pedersen 2014); lottery demand (Bali, Brown, Murray & Tang 2017).

Rule	Original (Frazzini & Pedersen (2014))	This factsheet
Signal	Shrunk beta from 1-year volatility and 5-year correlation	252-day beta vs the equal-weight universe (no shrinkage)
Portfolio	Rank-weighted legs above/below median beta, each scaled to beta 1	Same construction
Weighting	Rank weights	Rank weights

3. Data load

Universe	Prices	Membership	Names eligible (median)	Currency
US	Sharadar SEP ccloseadj (total return), 805 tickers incl. delisted	Point-in-time: the S&P 500 members at each month-end (Sharadar add/remove history; corrected 2 Oct 2026)	~503	USD
EU	Project1 yfinance cache, Adj Close (TR)	Point-in-time: union of 11 national blue-chip indices (CAC 40, DAX, AEX, IBEX 35, FTSE MIB, OMXS30, OMXC25, OMXH25, BEL 20, PSI-20, WIG20), quarterly Wikipedia snapshots 2012–2026; 83% of member-quarters priced	~238	local (EUR, SEK, DKK, PLN)
UK	Project1 cache, Close + pence dividends (Yahoo under-adjusts LSE ~100×)	Point-in-time: FTSE 100 + FTSE 250, quarterly snapshots; 75% priced	~242	GBP
DK	Project1 cache, Adj Close	Point-in-time: OMX Copenhagen 25, quarterly snapshots; 91% priced	~19	DKK
SCANDI	Project1 cache, Adj Close	Point-in-time: OMXC25 + OMXS30 + OMXH25 (Oslo has no snapshot history); 88% priced	~62	local (DKK, SEK, EUR)
World	US Sharadar + UK + EU above	Point-in-time union of the three; local-currency returns summed, not converted	~963	mixed

Membership comes from Project2's cache of dated Wikipedia index pages (the same parsing as `run_wiki_pit.py` / `run_eu_pit.py`), with a short list of verified ticker renames (e.g. `NZYM-B` → `NSIS-B`, `WDH` → `DEMANT`, `DAI` → `MBG`). A name is eligible from the day after the snapshot that lists it, once it has 60 days of prices. Member-quarters without a price are mostly delisted names that Yahoo no longer serves, so some survivorship remains; see §13.

- **Data gap:** Financing uses the USD risk-free rate for all markets, which overstates the cost in EU and Denmark while their rates were negative (2015–21). Betas are raw, not shrunk.

4. Signal creation

$BAB_{t+1} = r_L/\beta_L - r_H/\beta_H$, with rank-weighted low- and high-beta legs (the net long cash is financed at the USD risk-free rate + 0.5% a year; the high-beta leg pays a size-tiered borrow fee). Signals use only information up to month-end t and are traded in month $t+1$. Eligibility: in the point-in-time universe on any of the last five trading days of month t .

5. Model build

- Sort eligible stocks into equal-count groups (deciles ≥ 100 names, quintiles 50–99, terciles < 50 ; fixed per universe).
- **Long/short** = low-beta stocks minus high-beta stocks; **long-only** = low-beta stocks; benchmarks: equal-weight universe and size-weighted universe.
- Frazzini & Pedersen construction: betas ranked, each leg weighted by rank distance from the median and scaled to an ex-ante beta of 1; no risk-free rate is subtracted.
- Costs: 10 bp per side on the one-way turnover of each leg; short-leg borrow fee 0.25% / 0.75% / 2% a year by size tier (largest 50% / next 30% / smallest 20% of the universe, `code/borrow.py`); net long cash financed at the USD risk-free rate + 0.5%. Code: `code/xs.py` (engine), `code/analyse_xs.py`, `code/build_xs.py`.

6. Performance in detail

US (S&P 500 members, point-in-time), median 502 names, 10 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	+4.3%	+2.9%	16.8%	0.25 [-0.17, 0.70]	0.26	-33%	55%	1.17	0.14	+2.4% (0.6)	9.7%
Long-only (net)	+11.1%	+10.9%	11.8%	0.94 [0.54, 1.42]	1.48	-20%	66%	5.35	0.64	+2.6% (1.9)	7.0%
Low-beta stocks (gross)	+11.2%	+11.1%	11.8%	0.95 [0.56, 1.44]	1.51	-20%	66%	5.43	0.64	+2.8% (2.0)	7.0%
High-beta stocks (gross)	+15.7%	+14.0%	22.8%	0.69 [0.24, 1.22]	1.00	-39%	60%	3.08	1.39	-2.5% (-1.4)	13.0%
EW universe	+13.1%	+12.6%	15.5%	0.85 [0.41, 1.42]	1.29	-28%	67%	4.20	1.00	–	9.2%

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
Size-weighted universe	+14.9%	+14.9%	14.2%	1.05 [0.60, 1.61]	1.73	-24%	69%	5.29	0.86	+3.7% (2.4)	8.3%

EU (11 national blue-chip indices, point-in-time), median 264 names, 10 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	+4.6%	+3.8%	12.8%	0.36 [-0.10, 0.89]	0.44	-31%	55%	1.59	0.09	+3.6% (1.2)	8.0%
Long-only (net)	+9.3%	+9.1%	10.2%	0.91 [0.44, 1.47]	1.42	-18%	64%	4.33	0.62	+2.6% (2.3)	6.5%
Low-beta stocks (gross)	+9.5%	+9.3%	10.2%	0.93 [0.46, 1.50]	1.45	-18%	64%	4.43	0.62	+2.8% (2.4)	6.5%
High-beta stocks (gross)	+12.1%	+10.3%	21.4%	0.57 [0.12, 1.05]	0.79	-36%	56%	2.32	1.42	-3.2% (-2.2)	11.8%
EW universe	+10.8%	+10.2%	14.4%	0.75 [0.29, 1.30]	1.14	-26%	63%	3.20	1.00	–	8.4%
Size-weighted universe	+10.6%	+10.2%	13.7%	0.77 [0.32, 1.29]	1.21	-25%	58%	3.37	0.92	+0.8% (0.6)	7.9%

UK (FTSE 350, point-in-time), median 326 names, 10 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	+9.1%	+8.7%	11.7%	0.78 [0.22, 1.37]	1.21	-24%	58%	2.80	0.14	+7.7% (2.5)	7.4%
Long-only (net)	+9.6%	+9.6%	9.6%	1.01 [0.45, 1.68]	1.55	-19%	63%	4.25	0.62	+3.7% (3.8)	6.1%
Low-beta stocks (gross)	+9.9%	+9.8%	9.6%	1.03 [0.47, 1.71]	1.60	-19%	63%	4.35	0.62	+3.9% (4.0)	6.1%
High-beta stocks (gross)	+9.0%	+6.9%	21.3%	0.42 [-0.04, 0.95]	0.49	-42%	55%	1.71	1.44	-4.8% (-3.3)	12.6%
EW universe	+9.6%	+8.9%	14.2%	0.68 [0.19, 1.28]	0.95	-30%	58%	2.87	1.00	–	8.8%
Size-weighted universe	+8.4%	+7.8%	12.9%	0.65 [0.16, 1.25]	0.90	-28%	62%	2.68	0.84	+0.3% (0.2)	7.8%

DK (OMXC25, point-in-time), median 20 names, 3 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	+1.7%	+0.3%	16.5%	0.10 [-0.38, 0.56]	0.03	-34%	55%	0.39	0.04	+1.1% (0.3)	9.8%
Long-only (net)	+10.7%	+10.4%	12.4%	0.87 [0.40, 1.37]	1.39	-19%	60%	3.62	0.66	+2.4% (1.2)	6.9%

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
Low-beta stocks (gross)	+10.9%	+10.7%	12.4%	0.88 [0.42, 1.39]	1.43	-19%	60%	3.70	0.66	+2.6% (1.4)	6.9%
High-beta stocks (gross)	+15.8%	+14.2%	22.1%	0.72 [0.17, 1.36]	1.04	-49%	61%	2.62	1.32	-0.9% (-0.4)	13.3%
EW universe	+12.6%	+12.0%	15.2%	0.83 [0.31, 1.44]	1.27	-33%	66%	3.31	1.00	–	8.8%
Size-weighted universe	+11.1%	+10.2%	16.4%	0.68 [0.11, 1.33]	0.94	-40%	62%	2.55	0.93	-0.6% (-0.2)	10.7%

SCANDI (OMXC25 + OMXS30 + OMXH25, point-in-time), median 70 names, 5 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	+3.8%	+3.1%	12.5%	0.31 [-0.22, 0.82]	0.37	-26%	56%	1.23	0.04	+3.3% (1.1)	6.9%
Long-only (net)	+11.2%	+11.2%	10.6%	1.05 [0.57, 1.58]	1.83	-18%	63%	4.73	0.65	+3.0% (2.4)	5.8%
Low-beta stocks (gross)	+11.4%	+11.4%	10.6%	1.07 [0.59, 1.60]	1.87	-18%	63%	4.81	0.65	+3.2% (2.6)	5.8%
High-beta stocks (gross)	+14.2%	+12.7%	20.7%	0.69 [0.18, 1.31]	0.97	-43%	61%	2.66	1.39	-3.3% (-2.3)	12.6%
EW universe	+12.6%	+12.3%	14.2%	0.89 [0.37, 1.51]	1.38	-29%	66%	3.63	1.00	–	8.4%
Size-weighted universe	+11.4%	+11.0%	13.5%	0.84 [0.37, 1.41]	1.32	-26%	63%	3.59	0.90	-0.0% (-0.0)	8.1%

World (US + UK + EU, point-in-time), median 1088 names, 10 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	+5.9%	+5.1%	13.3%	0.44 [0.02, 0.93]	0.57	-23%	57%	2.12	0.24	+3.2% (1.1)	8.7%
Long-only (net)	+9.9%	+9.6%	11.6%	0.85 [0.41, 1.40]	1.28	-21%	66%	4.31	0.67	+2.4% (2.4)	7.2%
Low-beta stocks (gross)	+10.1%	+9.9%	11.6%	0.87 [0.43, 1.42]	1.31	-21%	67%	4.41	0.67	+2.6% (2.6)	7.2%
High-beta stocks (gross)	+12.7%	+10.6%	22.7%	0.56 [0.10, 1.10]	0.74	-41%	58%	2.25	1.37	-2.8% (-2.0)	13.0%
EW universe	+11.3%	+10.5%	16.0%	0.71 [0.25, 1.28]	1.02	-30%	61%	3.07	1.00	–	9.7%
Size-weighted universe	+11.0%	+10.3%	14.8%	0.74 [0.28, 1.31]	1.10	-28%	61%	3.19	0.91	+0.7% (0.8)	8.9%

Sharpe CI: stationary bootstrap (mean block 6 months, 5,000 draws). Max drawdown on monthly returns.

Calendar-year returns, L/S (net)

Year	US	EU	UK	DK	SCANDI	World
2013	+10.6%	+6.2%	+43.3%	-10.8%	+7.8%	+22.6%
2014	+17.3%	+21.5%	+7.9%	-1.5%	+24.1%	+16.0%
2015	+9.8%	+14.1%	+27.3%	-5.4%	+13.0%	+14.4%
2016	-4.5%	+2.1%	+8.6%	-12.7%	-22.5%	-9.1%
2017	+11.3%	+14.4%	+21.5%	+30.6%	+11.6%	+18.7%
2018	+3.2%	+7.8%	-8.0%	-17.9%	+7.0%	-3.3%
2019	+16.3%	+8.3%	+17.7%	+45.0%	+16.7%	+17.6%
2020	-14.1%	-7.6%	-0.6%	-25.9%	-8.6%	+0.1%
2021	+24.8%	+2.3%	+18.4%	+13.3%	+5.6%	+16.0%
2022	-3.5%	-15.0%	-9.4%	+8.2%	-0.9%	-12.6%
2023	-17.3%	-6.8%	-15.3%	-11.0%	-17.2%	-10.3%
2024	+10.1%	+2.7%	+11.4%	-3.7%	+2.1%	+11.4%
2025	-10.7%	+2.8%	+4.6%	+9.9%	+5.9%	-4.4%
2026	-3.4%	+5.0%	+5.2%	+7.8%	+8.2%	+1.7%

Calendar-year returns, long-only (net)

Year	US	EU	UK	DK	SCANDI	World
2013	+21.8%	+13.0%	+25.1%	+17.1%	+14.3%	+24.1%
2014	+18.9%	+10.6%	+5.8%	+11.2%	+18.7%	+9.5%
2015	+1.9%	+11.4%	+15.3%	+15.6%	+16.6%	+4.6%
2016	+9.5%	+10.9%	+15.8%	+2.3%	+10.6%	+2.2%
2017	+14.0%	+15.5%	+19.8%	+29.1%	+21.7%	+19.5%
2018	-1.4%	-2.0%	-8.4%	-13.3%	-0.9%	-7.0%
2019	+27.5%	+20.8%	+20.9%	+36.6%	+28.4%	+24.2%
2020	+8.4%	+2.6%	+0.9%	+10.0%	+9.9%	+10.6%
2021	+26.0%	+12.6%	+15.3%	+15.5%	+15.9%	+17.9%
2022	-4.6%	-9.7%	-11.1%	-5.0%	-9.1%	-10.9%
2023	+3.0%	+9.3%	+1.5%	+5.8%	+1.4%	+7.5%
2024	+13.7%	+6.6%	+11.5%	+2.0%	+4.1%	+11.4%
2025	+6.0%	+15.9%	+13.6%	+15.8%	+15.8%	+13.1%
2026	+8.7%	+10.3%	+11.1%	+7.8%	+10.1%	+10.4%

7. Trading record

The rebalance log per universe is in `results/xs_FS10_record_<universe>.csv` (names per leg, turnover, leg returns, gross and net L/S).

Universe	Rebalances	Names long / short	Turnover long leg / month	Turnover short leg / month	L/S months positive	Worst L/S month	When
US	163	250 / 250	13%	5%	55%	-13.8%	2021-02
EU	163	130 / 130	14%	5%	55%	-11.8%	2021-02
UK	163	162 / 162	17%	6%	58%	-11.0%	2020-11
DK	163	10 / 10	15%	6%	55%	-12.0%	2026-03
SCANDI	163	35 / 35	13%	6%	56%	-9.6%	2021-02
World	163	541 / 541	16%	6%	57%	-12.0%	2021-09

Current book (signal at 31 August 2026; first 8 names per leg)

Universe	Long: low-beta stocks	Short: high-beta stocks
US	CF, OXY, EOG, FANG, XOM, APA, CVX, DVN	MRNA, SMCI, BLDR, CCL, NCLH, COIN, UAL, HOOD
EU	REP.MC, NESTE.HE, GALP.LS, ENI.MI, TTE.PA, SHELL.AS, JSW.WA, KPN.AS	MTS.MC, MT.AS, IFX.DE, KGH.WA, ENR.DE, IAG.MC, BOL.ST, GLE.PA
UK	HBR.L, ITH.L, BP.L, SHEL.L, ENOG.L, BEZ.L, BHM.G.L, IMB.L	RPI.L, GDWN.L, RHIM.L, WIZZ.L, ANTO.L, FRES.L, HOC.L, PAF.L
DK	TRYG.CO, NSIS-B.CO, JYSK.CO, ISS.CO, NDA-DK.CO, CARL-B.CO, DANSKE.CO, RBREW.CO	ZEAL.CO, GN.CO, NOVO-B.CO, AMBU-B.CO, ROCK-B.CO, ORSTED.CO, DEMANT.CO, DSV.CO
SCANDI	NESTE.HE, ELISA.HE, TELIA.ST, TRYG.CO, TEL2-B.ST, EVO.ST, ESSITY-B.ST, MAERSK-B.CO	KCR.HE, METSO.HE, BOL.ST, ROCK-B.CO, ZEAL.CO, GN.CO, QTCOM.HE, KALMAR.HE

8. Risk and factor attribution

JKP 7 themes (US, UK, DK, World) or French Europe 5F + WML (EU, SCANDI); Newey-West t, 6 lags.

Long/short

Universe	Factor model	Alpha / yr	t(alpha)	R ²	Loadings with t ≥ 2 (t)
US	JKP USA 7 themes	-4.2%	-1.84	0.64	mkt +0.86 (+8.3), value -0.63 (-4.0), low_risk +1.72 (+10.3)

Universe	Factor model	Alpha / yr	t(alpha)	R ²	Loadings with t ≥ 2 (t)
EU	French Europe 5F + WML	-2.4%	-0.70	0.29	Mkt-RF +0.25 (+3.9), RMW +1.15 (+4.2), CMA +0.55 (+2.4), WML +0.33 (+2.8)
UK	JKP GBR 7 themes	+3.7%	1.13	0.50	mkt +0.39 (+7.5), momentum +0.25 (+2.9), low_risk +0.96 (+5.5), quality +0.62 (+4.2)
DK	JKP DNK 7 themes	-0.7%	-0.20	0.38	mkt +0.38 (+4.9), low_risk +1.16 (+7.6)
SCANDI	French Europe 5F + WML	-1.7%	-0.55	0.12	Mkt-RF +0.13 (+2.2), RMW +0.55 (+2.2), WML +0.33 (+3.5)
World	JKP World 7 themes	+1.1%	0.47	0.47	mkt +0.69 (+8.5), value -0.37 (-2.4), momentum +0.35 (+2.9), low_risk +1.24 (+8.5)

Long-only

Universe	Factor model	Alpha / yr	t(alpha)	R ²	Loadings with t ≥ 2 (t)
US	JKP USA 7 themes	+1.7%	1.59	0.87	mkt +0.91 (+30.4), size -0.23 (-3.3), value -0.14 (-2.4), low_risk +0.71 (+9.1)
EU	French Europe 5F + WML	+3.7%	1.94	0.65	Mkt-RF +0.54 (+15.0), RMW +0.46 (+3.3)
UK	JKP GBR 7 themes	+4.4%	2.27	0.69	mkt +0.45 (+13.5), quality +0.39 (+4.1)
DK	JKP DNK 7 themes	+5.3%	2.25	0.56	mkt +0.61 (+11.9), low_risk +0.25 (+2.8), quality +0.13 (+2.3)
SCANDI	French Europe 5F + WML	+6.3%	2.93	0.50	Mkt-RF +0.49 (+10.3), RMW +0.47 (+4.0)
World	JKP World 7 themes	+4.3%	2.79	0.78	mkt +0.70 (+15.7), low_risk +0.48 (+6.9)

9. Statistical verdict

- **Gate:** $|t| > 2.87$ (2 books $\times$ 6 universes, Bonferroni 0.05/12, two-sided).
- **L/S mean:** positive in all six; passes in 0 of 6.
- **L/S CAPM alpha:** +1.1% to +7.7%, $|t|$ past the gate in 0 of 6.
- **Long-only:** Sharpe above the equal-weight universe in all six; alpha +2.4% to +3.7%, passing in 1 of 6.
- **Probabilistic Sharpe** of the L/S, $P(\text{true Sharpe} > 0)$: US 0.82, EU 0.90, UK 1.00, DK 0.64, SCANDI 0.87, World 0.94.
- **Publication decay:** gross L/S +4.7% to +17.8% a year in 2013–19 against +3.2% to +6.9% in 2020–26.

10. What goes wrong, and how to fix it

Method. Diagnose on the design window (2013–2019); apply the same pre-declared fix ladder used for every sort strategy in FS03–FS12; judge on the holdout (Jan 2020 – Aug 2026) with a pooled six-universe paired bootstrap of the Sharpe difference against the previous step. 25 fix trials across FS03–FS12, gate $p < 0.0020$.

10.1 Diagnosis (design window 2013–2019)

Universe	L/S gross / yr	Beta	Beta drag / yr	CAPM alpha (gross)	Costs / yr	Long leg vs EW / yr	EW vs short leg / yr	Turnover / month	Worst months
US	+11.6%	-0.08	-1.0%	+12.6%	-0.4%	-0.4%	+0.3%	18%	2013-05 -7%, 2016-04 -7%, 2016-11 -6%
EU	+12.7%	-0.02	-0.2%	+12.9%	-0.4%	+0.7%	+0.6%	19%	2019-04 -5%, 2016-10 -5%, 2016-04 -4%
UK	+17.8%	0.38	+4.4%	+13.4%	-0.6%	+1.4%	+0.7%	24%	2016-11 -7%, 2019-04 -4%, 2018-12 -4%
DK	+4.7%	0.20	+3.0%	+1.6%	-0.5%	-2.0%	-4.9%	22%	2013-05 -10%, 2018-10 -10%, 2016-10 -9%
SCANDI	+10.0%	-0.01	-0.1%	+10.2%	-0.5%	-0.1%	-2.2%	20%	2016-11 -8%, 2019-06 -7%, 2016-10 -6%
World	+12.8%	0.11	+1.2%	+11.6%	-0.5%	-0.3%	+0.7%	23%	2018-05 -5%, 2018-02 -5%, 2016-04 -5%

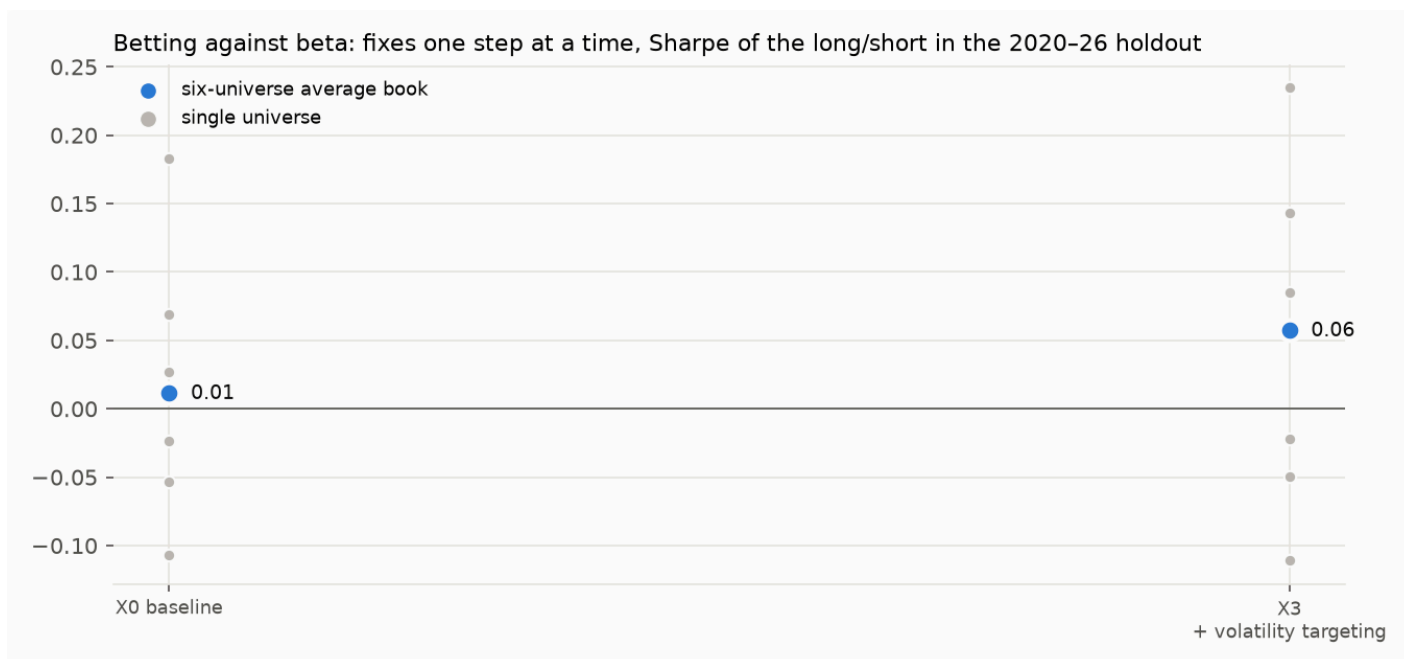
On average across the six universes the gross spread was +11.6% a year, of which the market exposure (beta +0.10) contributed +1.2%; the beta-adjusted spread (CAPM alpha) was +10.4%. Costs took 0.5% a year at 21% monthly turnover across both legs. The long leg beat the universe by -0.1% and the short leg lagged it by -0.8% a year, so most of the spread comes from the long side. Holding longer does not change the picture much: the gross spread 2, 3 and 6 months after formation averages +3.4%, +4.0%, +4.9% a year.

10.2 The fix ladder

X0 baseline → X3 volatility targeting (BAB is beta-neutral and rank-weighted by construction, so the beta-neutral and buffer steps do not apply).

Step	Design Sharpe (avg)	Δ design (p)	Holdout Sharpe (book)	Δ holdout (p)	Universes improved	Holdout return / yr	Holdout max DD	Holdout alpha vs EW (t)	Costs / yr
X0 baseline	0.90	–	0.01	–	–	+0.2%	-22%	-1.2% (-0.3)	0.5%
X3 + volatility targeting	0.75	-0.03 (p 0.518)	0.06	+0.05 (p 0.382)	4 of 6	+0.6%	-22%	-0.4% (-0.1)	0.5%

✓ = passes the gate in the holdout. Design Sharpe = average across universes; tests and holdout columns use the six-universe equal-weighted book.

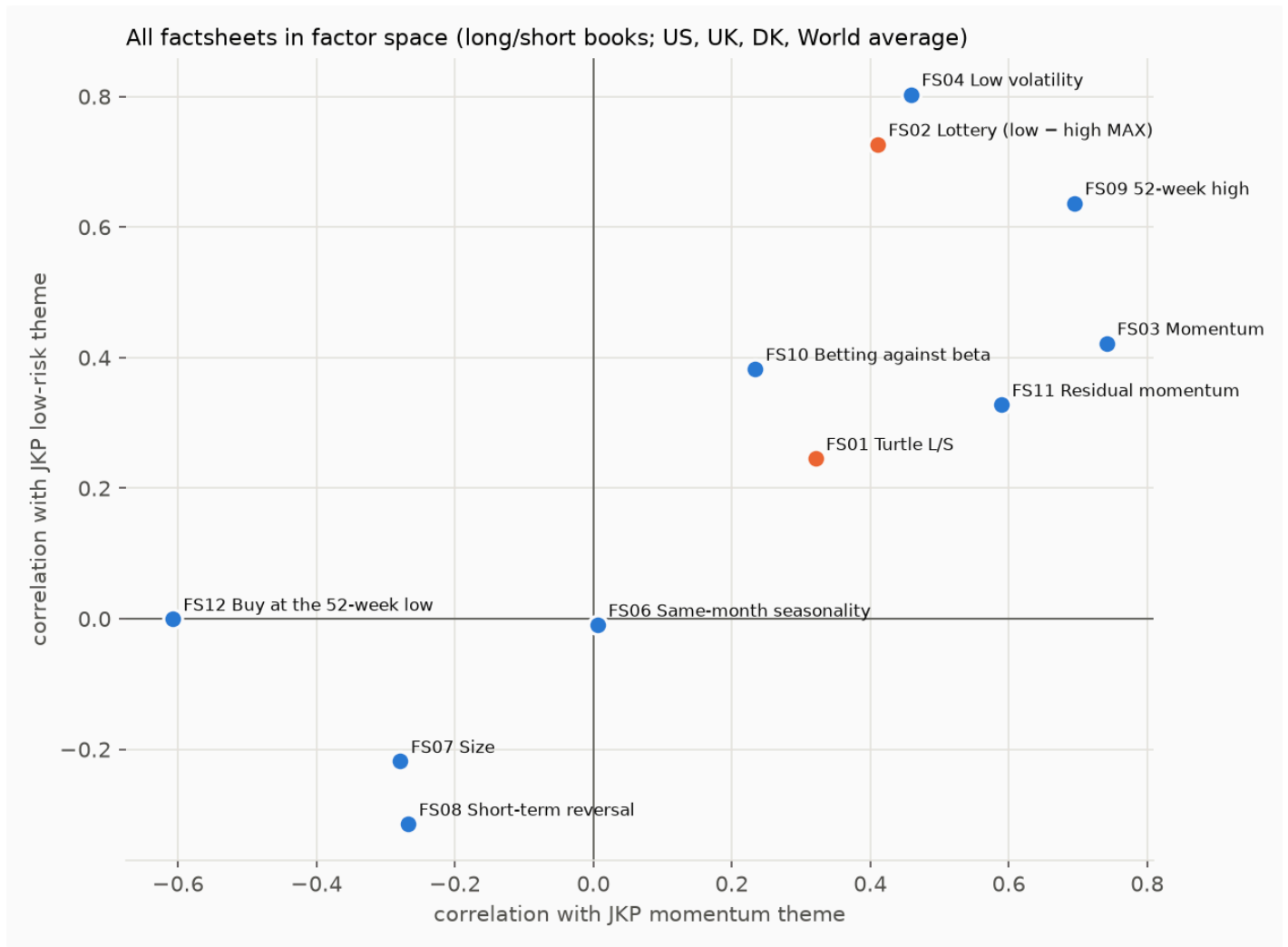


- **X3 + volatility targeting:** -0.03 design, +0.05 holdout; helps in one window and hurts in the other: not reliable.

Where it ends: holdout Sharpe 0.01 → 0.06, return +0.2% → +0.6% a year (six-universe book).

11. Classification: where does it fit?

Dimension	Betting against beta
Series family	Low risk
Academic style	Defensive / betting against beta
Signal	Price only (daily total returns)
Cross-section vs time series	Cross-sectional: stocks ranked against each other each month
Horizon and turnover	One month; 21% of the two legs replaced monthly
Market exposure	L/S beta 0.04 to 0.24
Payoff shape	Treynor–Mazuy γ -2.38 to -0.58 (t -5.0 to -0.6); worst 10% of market months -3.4% to -0.1% a month, best 10% -1.0% to +0.7%
Nearest factor theme	size (1 of 6 universes), RMW (1 of 6 universes)
Nearest library signals (returns)	Volatility (252 days) (6 universes), Volatility (6 universes), Range (MAX–MIN) (5 universes)
Economic rationale	Leverage-constrained investors buy high-beta stocks instead of leveraging the market, bidding them up (Black 1972; Frazzini & Pedersen 2014); lottery demand (Bali, Brown, Murray & Tang 2017).
Publication	Black, Jensen & Scholes 1972; Frazzini & Pedersen 2014; critique by Novy-Marx & Velikov (2022) on construction and costs
Where it fits	In the low risk cluster of the library; with some information beyond its neighbours.

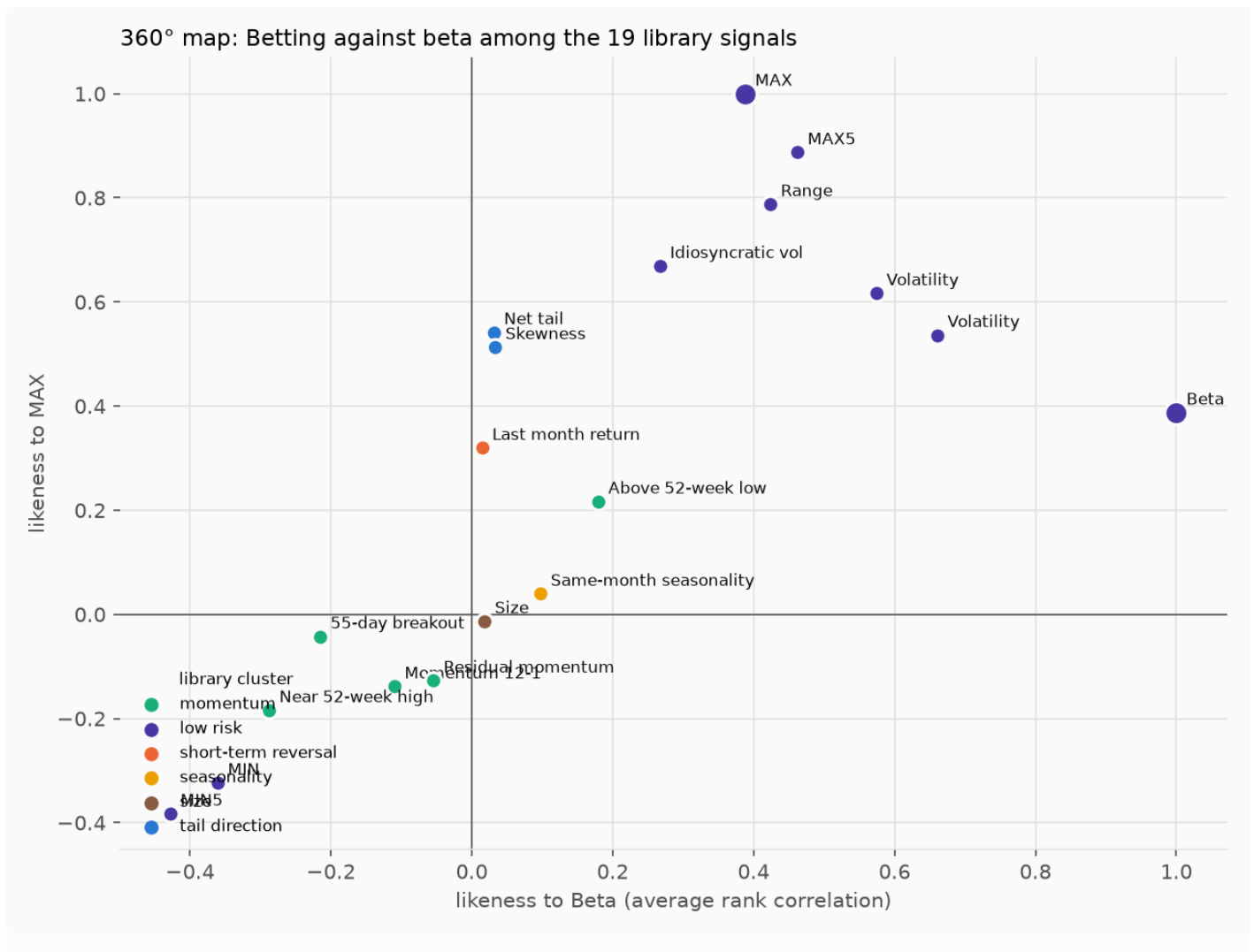


12. 360° view

The signal among the 19 price signals of the shared library (`code/library.py`), on the same universes and months (equal-weighted top minus bottom quantile, gross). Its natural mirror here is **MAX**.

Signal (top minus bottom, raw sign)	Cluster	Likeness: stocks	Likeness: returns	Raw L/S, avg of 6	CAPM alpha range	Universes alpha $t \geq +2 / \leq -2$
Beta (this strategy)	low risk	+1.00	+1.00	+4.4%	-12.6% to -4.2%	0 / 4
Volatility (252 days)	low risk	+0.66	+0.86	+2.4%	-12.9% to -4.8%	0 / 3
Volatility	low risk	+0.57	+0.85	+1.1%	-13.7% to -7.9%	0 / 5
Range (MAX–MIN)	low risk	+0.42	+0.77	-0.7%	-15.4% to -4.8%	0 / 4
MIN5	low risk	-0.43	-0.75	-0.9%	+2.3% to +13.7%	4 / 0
MAX5	low risk	+0.46	+0.73	+0.6%	-11.3% to -3.5%	0 / 2

Signal (top minus bottom, raw sign)	Cluster	Likeness: stocks	Likeness: returns	Raw L/S, avg of 6	CAPM alpha range	Universes alpha $t \geq +2$ / ≤ -2
MAX (best day) (mirror)	low risk	+0.39	+0.72	+0.1%	-13.3% to -2.0%	0 / 2
MIN (worst day)	low risk	-0.36	-0.71	-0.7%	+3.3% to +11.1%	4 / 0
Near 52-week high	momentum	-0.29	-0.69	+3.7%	+9.3% to +22.3%	6 / 0
Idiosyncratic vol	low risk	+0.27	+0.61	-3.6%	-15.5% to -4.8%	0 / 4
55-day breakout	momentum	-0.21	-0.60	+1.4%	+4.1% to +13.0%	4 / 0
Momentum 12-1	momentum	-0.11	-0.47	+7.2%	+4.3% to +19.4%	5 / 0
Size	size	+0.02	-0.28	+0.1%	-3.8% to +9.9%	1 / 0
Residual momentum	momentum	-0.05	-0.28	+6.4%	+8.0% to +12.2%	6 / 0
Last month return	short-term reversal	+0.01	-0.27	+1.8%	-3.2% to +11.0%	2 / 0
Skewness	tail direction	+0.03	+0.10	-0.4%	-6.7% to +3.9%	0 / 2
Same-month seasonality	seasonality	+0.10	-0.09	-1.9%	-9.4% to +6.6%	0 / 1
Net tail (MAX+MIN)	tail direction	+0.03	-0.02	-0.9%	-5.2% to +5.2%	0 / 0
Above 52-week low	momentum	+0.18	-0.01	+7.5%	+5.2% to +13.6%	4 / 0



Spanning. Regressing the L/S on its five closest library neighbours (by return correlation) and the market:

Universe	Alpha after neighbours / yr	t	R ²	Neighbours with t ≥ 2
US	-0.3%	-0.13	0.88	Volatility +1.01 (+7.2), Range -0.51 (-2.6)
EU	-1.9%	-0.54	0.73	Volatility +0.45 (+2.6), Volatility +0.41 (+2.2)
UK	-5.9%	-2.19	0.84	Volatility +0.43 (+3.7), Near 52-week high -0.17 (-2.2)
DK	+4.3%	1.43	0.78	Volatility +0.71 (+11.5), Volatility +0.18 (+2.7)
SCANDI	-2.6%	-0.90	0.68	
World	-0.7%	-0.37	0.90	Volatility +0.85 (+7.4)

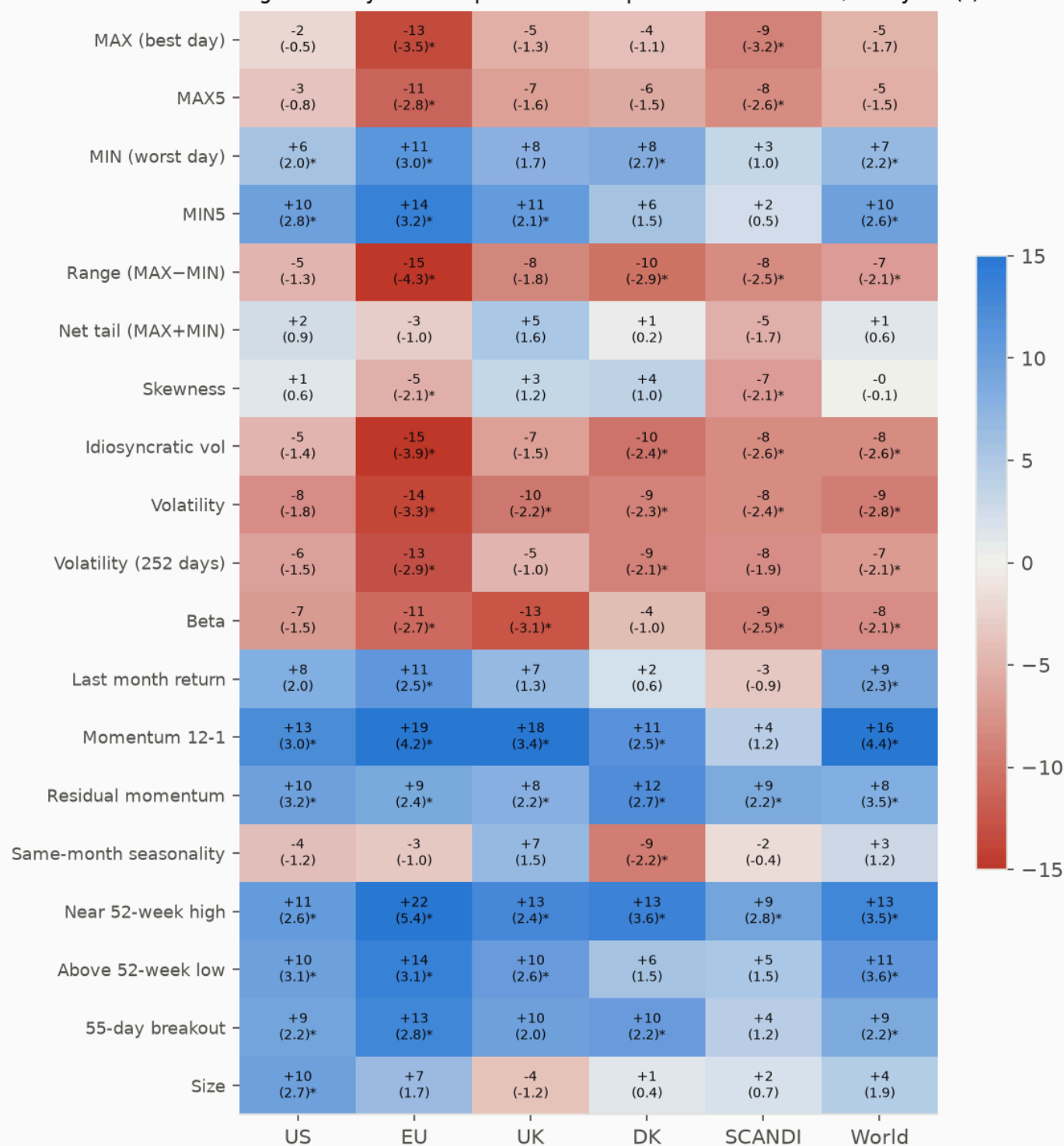
3×3 sort, next-month return a year
(US, EU, UK, SCANDI, World averaged)

	low MAX	middle	high MAX
low Beta	10.5%	10.2%	10.2%
middle	11.8%	12.1%	9.2%
high Beta	14.0%	13.7%	11.7%

Holding MAX fixed (down a column), moving from low to high Beta changes the return by +3.5, +3.5 and +1.5 points a year; holding Beta fixed (along a row), moving from low to high MAX changes it by -0.3, -2.7 and -2.2. **Market direction.** Top-minus-bottom return in rising vs falling market months (six-universe average): +32% vs -45% a year. **Horizon:** the gross spread at 1, 2, 3 and 6 months after formation averages +4.4%, +3.4%, +4.0%, +4.9% a year.

The full library, with the CAPM alpha of every signal in every universe:

Signal library: CAPM alpha of each top-minus-bottom sort, % a year (t)



13. Caveats

1. **Residual survivorship.** Point-in-time membership everywhere, but 9–25% of member-quarters outside the US have no price (mostly delisted names); long books are flattered and short books penalised by an unknown amount.
2. **Currency.** Local currency; World sums local-currency returns without conversion.

3. **Equal weighting and flat trading costs.** 10 bp per side for every stock and a size-tiered borrow fee, but no market impact; blue-chip universes only.
4. **Data gap.** Financing uses the USD risk-free rate for all markets, which overstates the cost in EU and Denmark while their rates were negative (2015–21). Betas are raw, not shrunk.
5. **Not a registered trial.** Descriptive; the fix ladder is counted inside FSO3–FS12 (25 trials).

14. Academic references

- Frazzini, A. & Pedersen, L. H. (2014). Betting against beta. *Journal of Financial Economics*, 111(1), 1–25.
- Black, F. (1972). Capital market equilibrium with restricted borrowing. *Journal of Business*, 45(3), 444–455.
- Bali, T. G., Brown, S., Murray, S. & Tang, Y. (2017). A lottery-demand-based explanation of the beta anomaly. *Journal of Financial and Quantitative Analysis*, 52(6), 2369–2397.
- Novy-Marx, R. & Velikov, M. (2022). Betting against betting against beta. *Journal of Financial Economics*, 143(1), 80–106.
- Newey, W. & West, K. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.
- Politis, D. & Romano, J. (1994). The stationary bootstrap. *Journal of the American Statistical Association*, 89(428), 1303–1313.
- Bailey, D. & López de Prado, M. (2012). The Sharpe ratio efficient frontier. *Journal of Risk*, 15(2), 3–44.
- Jensen, T. I., Kelly, B. & Pedersen, L. H. (2023). Is there a replication crisis in finance? *Journal of Finance*, 78(5), 2465–2518.
- Fama, E. & French, K. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1), 1–22.
- McLean, R. D. & Pontiff, J. (2016). Does academic research destroy stock return predictability? *Journal of Finance*, 71(1), 5–32.
- Novy-Marx, R. & Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1), 104–147.
- Barroso, P. & Santa-Clara, P. (2015). Momentum has its moments. *Journal of Financial Economics*, 116(1), 111–120.
- Moreira, A. & Muir, T. (2017). Volatility-managed portfolios. *Journal of Finance*, 72(4), 1611–1644.
- Treynor, J. & Mazuy, K. (1966). Can mutual funds outguess the market? *Harvard Business Review*, 44(4), 131–136.

15. Reproduce

`code/data.py` → `signals.py`, `signals2.py`, `library.py` → `xs.py` + `analyse_xs.py` FS10 → `build_xs.py` FS10.

Metrics use Project1 `InvestmentLibrary`. Licensed and cached prices are not included.